

Experiment 7.3

Determination of Capacitance per Unit Length of a Coaxial Cable

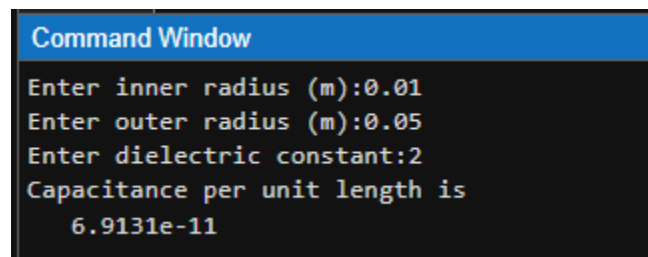
Code File Name: Exp7_3_CoaxialCableCapacitance.m

Code

```
clc;
epsilon0 = 8.854e-12;
a = input('Enter inner radius (m):');
b = input('Enter outer radius (m):');
er = input('Enter dielectric constant:');
epsilon = epsilon0*er;
C = (2*pi*epsilon)/log(b/a);
disp('Capacitance per unit length is');
disp(C);
```

Command Prompt

```
Enter inner radius (m):0.01
Enter outer radius (m):0.05
Enter dielectric constant:2
Capacitance per unit length is
    6.9131e-11
```

A screenshot of a MATLAB Command Window. The title bar is blue and says "Command Window". The background is black with white text. The text shows the prompts and inputs from the previous block, followed by the output of the capacitance calculation.

```
Command Window
Enter inner radius (m):0.01
Enter outer radius (m):0.05
Enter dielectric constant:2
Capacitance per unit length is
    6.9131e-11
```